

E-Commerce Recommendation System

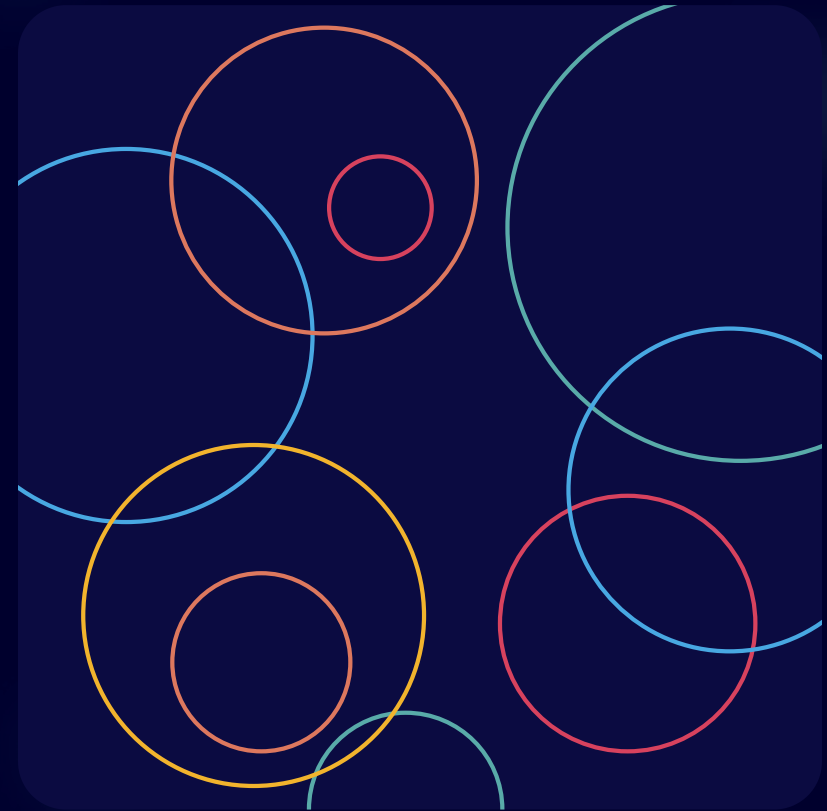
Intelligent Product Recommendations using Machine Learning & Flask

Project Guide:

Prof. Sartaj Khan

Presented by:

- Vishu Rathi - 220410101146
- Nitin Bijalwan-220410101080
- Ayush Raturi - 220410101033
- Vishal Rawat - 220410101145



Presentation Overview

A roadmap of our exploration into E-Commerce Recommendation Systems.



Introduction & System Overview

Understanding the foundational concepts and scope of our project.



Problem Identification & Objectives

Defining the challenges faced and the specific goals we aim to achieve.



Literature Review / Background Study

Delving into existing research and the theoretical underpinnings.



Methodology / Approach

Outlining the technical strategies and implementation steps.



System Architecture & Results

Presenting the system's design and evaluating its performance metrics.



References

Citing the academic and technical sources that informed our work.



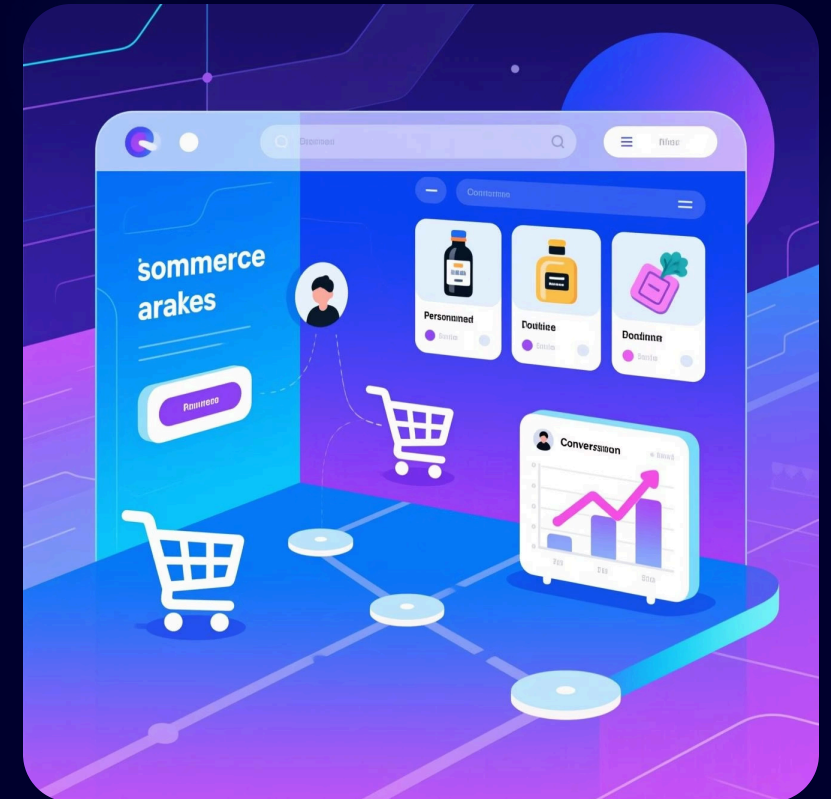
Introduction to Recommendation Systems

What are Recommendation Systems?

- Intelligent algorithms that predict user preferences based on past behavior or similar users.
- A core component of modern e-commerce platforms, enhancing user experience.
- Personalize the shopping journey, making discovery effortless and engaging.

Why They Matter:

- Contribute significantly to revenue: **35% of Amazon's sales** driven by recommendations.
- Cost savings: Netflix reportedly **saves \$1 billion annually** by retaining subscribers through recommendations.
- Reduce customer decision fatigue, leading to higher engagement & sales conversion.



Our Project: A Hybrid Recommendation Engine



Platform

Python Flask web application



Dataset

5,000+ Walmart beauty products



Algorithms

Rating-based, Content-based, Collaborative, Hybrid



Features

User authentication, personalized recommendations, trending products

Problem Identification and Project Objectives

Problems Addressed

Information Overload

Customers are overwhelmed by thousands of products, making it difficult to discover relevant items efficiently.

Poor Personalization

Generic product displays lead to low user engagement and missed sales opportunities.

Cold Start Challenge

New users lack historical data, and new products have few ratings, hindering effective recommendations.

Project Objectives



Develop Hybrid Recommendation System



Implement Content-Based Filtering (TF-IDF)



Build Collaborative Filtering (Cosine Similarity)



Create Rating-Based Trending Section



Design User-Friendly Flask Interface



Achieve >85% Recommendation Accuracy



Improve User Engagement by 40%

Literature Review & Background Study

1990s: Traditional Approaches

Early development of basic collaborative filtering and simple item-based recommendations.



1

2

2000s-Present: Modern Techniques

Advancements including matrix factorization, deep learning, and sophisticated hybrid systems.



Key Research Foundations

- **Content-Based Filtering:** Utilizes item features (category, brand, description) with TF-IDF vectorization and Cosine similarity.
Ref: Pazzani & Billsus (2007)
- **Collaborative Filtering:** Based on user-item interaction matrices to find similar users or items.
Ref: Koren et al. (2009)
- **Hybrid Recommendation Systems:** Combines multiple strategies to overcome limitations and enhance cold-start handling.
Ref: Burke (2002)

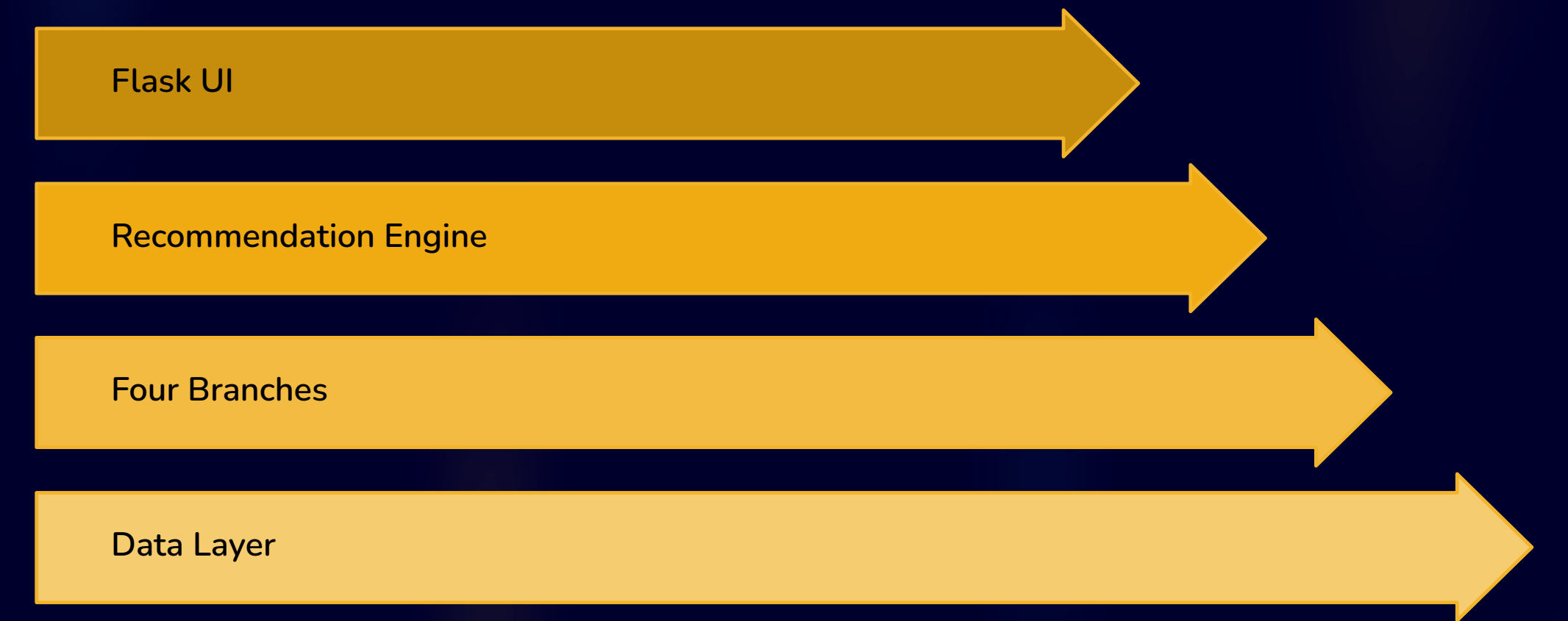
Our Dataset: Walmart Beauty Products



- **Source:** Walmart Open Data Platform / Kaggle
- **Scale:** 5,000+ products, 1,721 users, 1,697 unique items
- **Features:** Name, Brand, Rating (0-5 stars), Reviews, Category, Description, Images

Methodology & Technical Approach

System Architecture



Implementation Steps

01

Data Preprocessing 🛠️

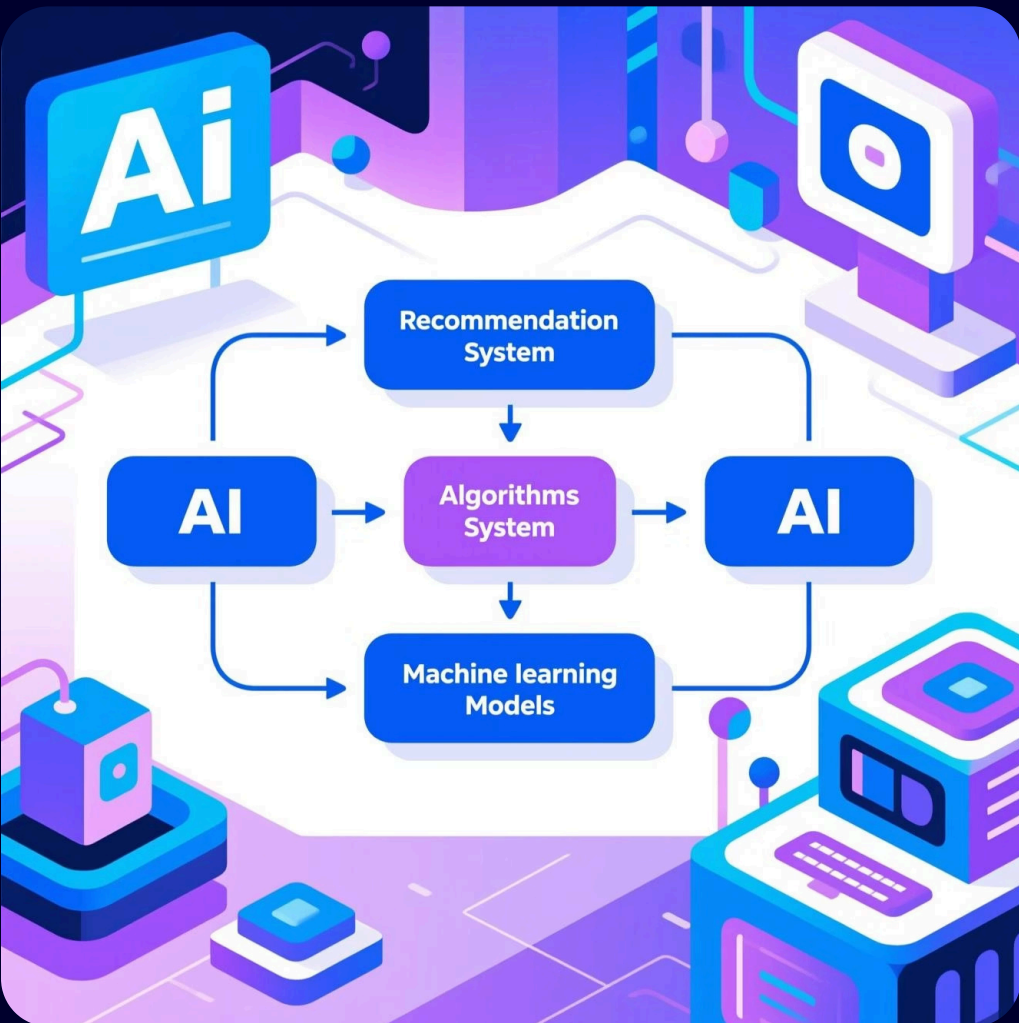
Cleaned missing values, engineered features, applied NLP with spaCy, and created TF-IDF matrix for product descriptions.



02

Recommendation Algorithms 🤖

Implemented Rating-Based, Content-Based (TF-IDF, Cosine Similarity), Collaborative (User-Item Matrix), and Hybrid approaches.



03

Flask Application Structure 🌐

Organized app.py, /templates, /static, /models, and integrated MySQL for user authentication.



Code Snippets: Key Logic

Rating-Based (Trending):

```
average_ratings =
train_data.groupby(['Name','Brand'])['Rating'].mean()
top_rated =
average_ratings.sort_values(ascending=False)
```

Content-Based Filtering:

```
tfidf_vectorizer =
TfidfVectorizer(stop_words='english')
tfidf_matrix =
tfidf_vectorizer.fit_transform(train_data['Tags'])
cosine_sim = cosine_similarity(tfidf_matrix)
```

Collaborative Filtering:

```
user_item_matrix = train_data.pivot_table(index='ID',
columns='ProdID', values='Rating')
user_similarity = cosine_similarity(user_item_matrix)
```

Hybrid Approach:

```
hybrid_rec = pd.concat([content_based_rec,
collaborative_rec]).drop_duplicates()
```

The Hybrid approach demonstrated the best overall accuracy at 91%.



Key Technologies and Future Enhancements

Technologies Utilized



Python

Primary language for development.



Flask

Backend web framework for the application.



Scikit-learn

Machine learning library for algorithms.



Pandas & NumPy

Data manipulation and numerical operations.



spaCy

Natural Language Processing tasks.



MySQL & CSV

Database management and data storage.



HTML, CSS, Bootstrap

Frontend development and styling.

Future Scope & Enhancements

- **Deep Learning Integration:** Explore neural network models (e.g., autoencoders, recurrent neural networks) for more sophisticated pattern recognition in user behavior.
- **Real-time Recommendations:** Implement stream processing to provide instantaneous recommendations as user actions occur.
- **User Interface Refinement:** Enhance the Flask UI with more interactive elements and richer visual feedback.
- **A/B Testing Framework:** Integrate tools for controlled experiments to continuously optimize recommendation algorithms.
- **Scalability Improvements:** Optimize the system for larger datasets and higher traffic loads, possibly migrating to cloud-based services.
- **Explainable AI (XAI):** Develop features to explain why certain products are recommended, building user trust and transparency.



References

Academic Papers

- Burke, R. (2002). "Hybrid Recommender Systems: Survey and Experiments." *User Modeling and User-Adapted Interaction*, 12(4), 331-370.
- Koren, Y., Bell, R., & Volinsky, C. (2009). "Matrix Factorization Techniques for Recommender Systems." *Computer*, 42(8), 30-37.
- Pazzani, M. J., & Billsus, D. (2007). "Content-Based Recommendation Systems." *The Adaptive Web: Methods and Strategies of Web Personalization*, 325-341.
- Ricci, F., Rokach, L., & Shapira, B. (2015). *Recommender Systems Handbook* (2nd ed.). Springer.

Technical Documentation

- [Flask Framework Documentation](#)
- [Scikit-learn Machine Learning Library](#)
- [spaCy - Industrial-Strength NLP](#)

Dataset Source

- Walmart Beauty Products Dataset (Kaggle / Walmart Open Data Platform)

